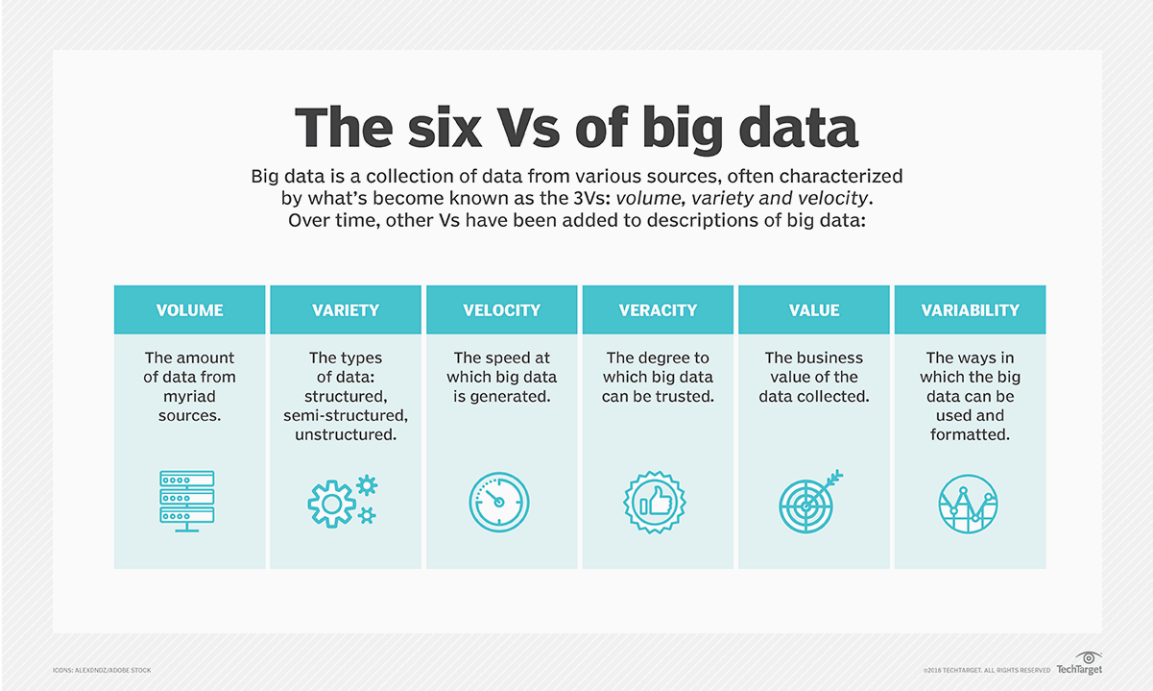
EL9: Big data

Big data



Biggish data

My definition:

- Only tabular data
- Original data might fit into physical RAM but ...
- might be multiplied due to temporary necessity
- physical RAM might be constrained due to other processes/settings
- things might just take too much time ... and life is short ...

Software

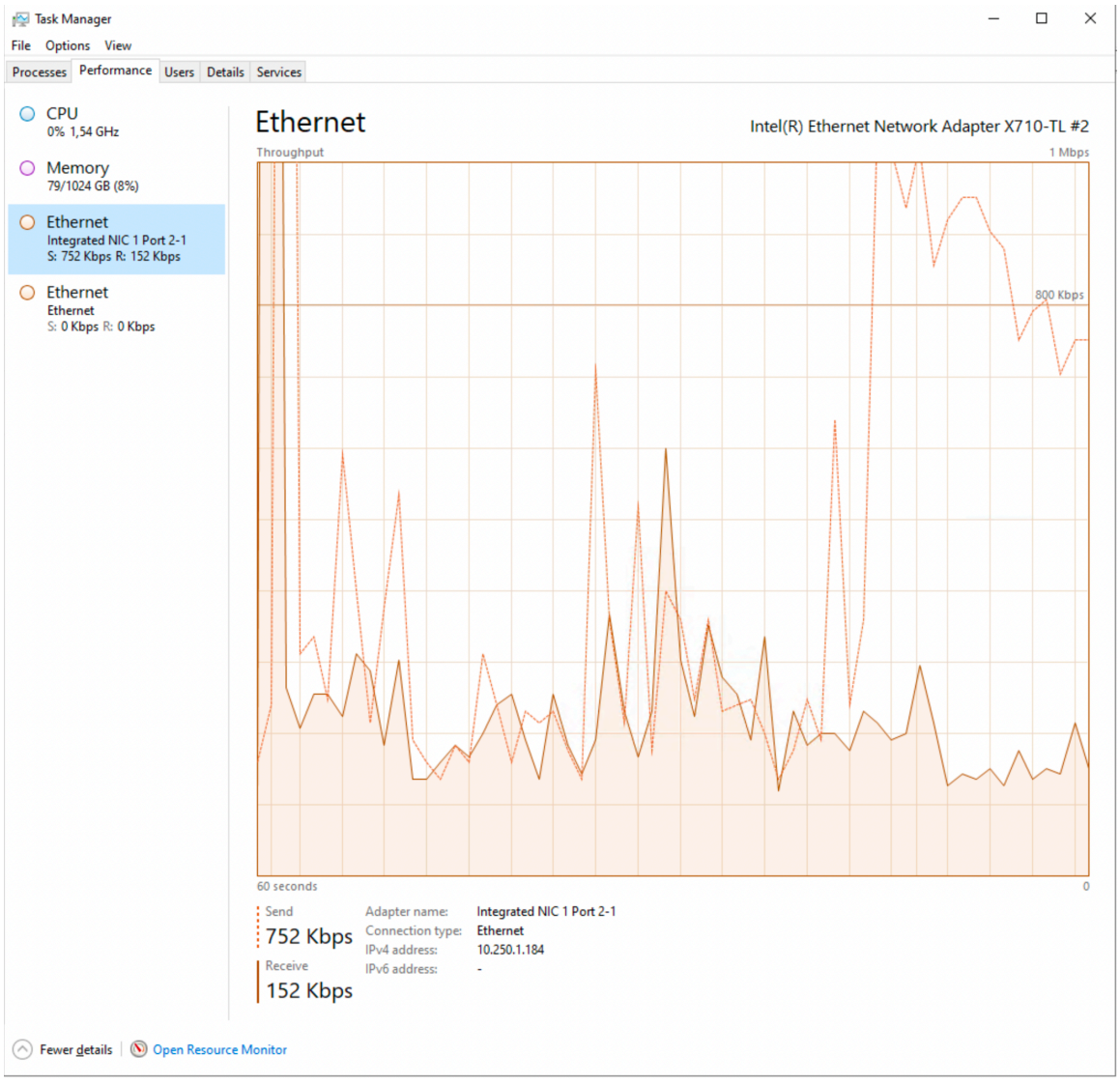
- SAS for big data files from register holders
- R for everything else
- Database?

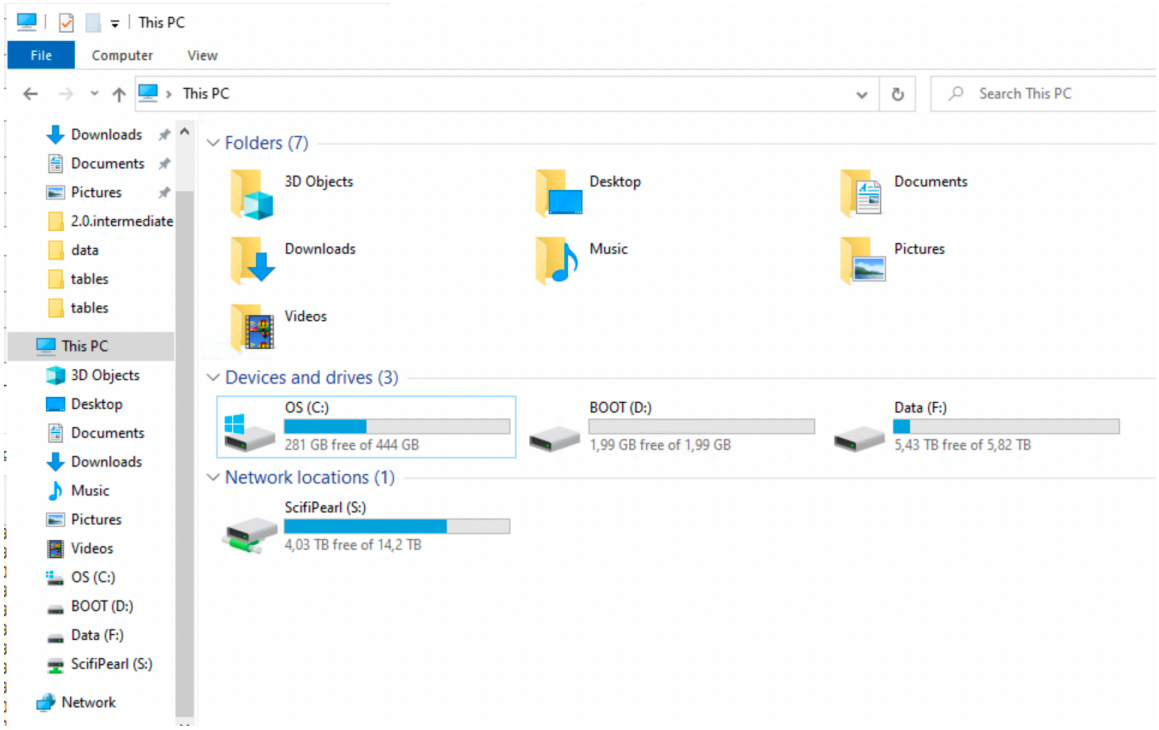
File format

File storage

- Avoid reading big files over ethernet connection

    - data stored on network drive
- Usually faster to first copy file to local SSD drive
    - `fs::file_copy(path, new_path)`
    - (*"libuv provides a wide variety of cross-platform sync and async file system operations."*)
    - obviously only if "local drive" is also in the (same) secure environment
- work with the project on disk
- move back
- Although not always possble with very large files it seems




This PC
File
Computer
View
This PC
Search This PC
Downloads
Documents
Pictures
2.0.intermediate
data
tables
tables
This PC
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
OS (C:)
BOOT (D:)
Data (F:)
ScifiPearl (S:)
Network
Folders (7)
3D Objects
Desktop
Documents
Downloads
Music
Pictures
Videos
Devices and drives (3)
OS (C:)
281 GB free of 444 GB
BOOT (D:)
1,99 GB free of 1,99 GB
Data (F:)
5,43 TB free of 5,82 TB
Network locations (1)
ScifiPearl (S:)
4,03 TB free of 14,2 TB

Comparison vs base equivalents

fs functions smooth over some of the idiosyncrasies of file handling with base R functions:

- Vectorization. All **fs** functions are vectorized, accepting multiple paths as input. Base functions are inconsistently vectorized.
- Predictable return values that always convey a path. All **fs** functions return a character vector of paths, a named integer or a logical vector, where the names give the paths. Base return values are more varied: they are often logical or contain error codes which require downstream processing.
- Explicit failure. If **fs** operations fail, they throw an error. Base functions tend to generate a warning and a system dependent error code. This makes it easy to miss a failure.
- UTF-8 all the things. **fs** functions always convert input paths to UTF-8 and return results as UTF-8. This gives you path encoding consistency across OSes. Base functions rely on the native system encoding.
- Naming convention. **fs** functions use a consistent naming convention. Because base R's functions were gradually added over time there are a number of different conventions used (e.g. `path.expand()` vs `normalizePath()`; `Sys.chmod()` vs `file.access()`).

`sync_project()`

- I have a function `sync_project()` in my `.Rprofile`
  - `usethis::edit_r_profile()`
- However, I don't include all data files in each individual project
- "Chek out" all files from network storage to disk
- Work with it
- Sync back to allow back-ups and common folder structure
- Git-repo kept for the network storage

```
sync_project <- function (from, to)
{
    loc_dirs <- c("S:/Projects/", "F:/xbuler/")
    loc_dirs_exp <- paste(loc_dirs, collapse = "|")
    proj <- gsub(loc_dirs_exp, "", getwd())
    locations <- paste0(loc_dirs, proj)
    locations_exp <- paste(paste0(locations, "/"), collapse = "|")
    stopifnot(getwd() %in% locations)
```

```
    from <- if (missing(from))
        getwd()
    to <- if (missing(to))
        setdiff(locations, from)
    ans <- askYesNo(paste("Sync from:\n ", from, "\n To: \n",
        to))
    stopifnot(ans)
    if (!fs::dir_exists(to)) {
        fs::dir_create(to)
        message("Target folder did not exist but has been created!")
    }
    fr <- dplyr::mutate(dplyr::select(fs::dir_info(from, all = TRUE,
        recurse = TRUE, regexp = ".git", invert = TRUE), path,
        type, size, modification_time), file = gsub(locations_exp,
        "", path), path_to = paste(to, file, sep = "/"))
    t <- dplyr::mutate(dplyr::select(fs::dir_info(to, all = TRUE,
        recurse = TRUE), path, type, size, modification_time),
        file = gsub(locations_exp, "", path))
    new <- dplyr::anti_join(fr, t, "file")
    new_dirs <- dplyr::filter(new, type == "directory")
    if (nrow(new_dirs)) {
        print(dplyr::select(new_dirs, file), n = Inf)
        ans <- askYesNo(paste(nrow(new_dirs), " dir(s) not in 'to'. Should I
create these?"))
        if (ans)
            fs::dir_create(new_dirs$path_to)
    }
    new_files <- dplyr::filter(new, type == "file")
    if (nrow(new_files)) {
        print(dplyr::select(new_files, file), n = Inf)
        ans <- askYesNo(paste(nrow(new_files), " file(s) not in 'to'. Should I
copy them?"))
        if (ans)
            purrr::walk2(new_files$path, new_files$path_to, fs::file_copy,
                .progress = TRUE)
    }
    diff_mod <- dplyr::filter(dplyr::left_join(fr, t, "file",
        suffix = c("_from", "_to")), type_from == "file",
modification_time_to !=
        modification_time_from)
    if (nrow(diff_mod)) {
        print(dplyr::select(diff_mod, file,
tidyselect::matches("modification_time|size")),
            n = Inf)
        ans <- askYesNo(paste(nrow(diff_mod), " file(s) have been updated.
Should I replace the older version?"))
        if (ans) {
            purrr::walk2(diff_mod$file, diff_mod$path_to, fs::file_copy,
```

```
                overwrite = TRUE, .progress = TRUE)
        }
    }
    dlt <- dplyr::filter(dplyr::anti_join(t, fr, "file"), !startsWith(file,
        ".git"))
    if (nrow(dlt)) {
        print(dlt, n = Inf)
        ans <- askYesNo(paste(nrow(dlt), " file(s) have been deleted in
'from'. Should I delete them in 'to'?"))
        if (ans) {
            purrr::walk(dlt$path, fs::file_delete, .progress = TRUE)
        }
    }
}
```

👎 Reading SAS files in R

- Common to get large register files in SAS-format (SoS, SCB)
- 👍 SAS can handle that (files on disk) 😀
- 👎 I am a terrible SAS user 😕
- 👎 Seems to be no easy way to read only parts of SAS-files into R directly
- `haven::read_sas()`?

`haven::read_sas()`

- 👍 May work if big enough RAM ...
- 👎 Whole file is first read into RAM (`readr::datasource(data_file)`)
- 👎 Most time spent on `df_parse_sas_file()`
- (Same for .dta-files [Stata])

```
haven::read_sas
```

```
function (data_file, catalog_file = NULL, encoding = NULL, catalog_encoding =
encoding,
    col_select = NULL, skip = 0L, n_max = Inf, cols_only = deprecated(),
    .name_repair = "unique")
{
    if (lifecycle::is_present(cols_only)) {
        lifecycle::deprecate_warn("2.2.0", "read_sas(cols_only)",
            "read_sas(col_select)")
        stopifnot(is.character(cols_only))
        col_select <- quo(c(!!!cols_only))
    }
    else {
        col_select <- enquo(col_select)
```

```
    }
    if (is.null(encoding)) {
        encoding <- ""
    }
    cols_skip <- skip_cols(read_sas, !!col_select, data_file,
        encoding = encoding)
    n_max <- validate_n_max(n_max)
    spec_data <- readr::datasource(data_file)
    if (is.null(catalog_file)) {
        spec_cat <- list()
    }
    else {
        spec_cat <- readr::datasource(catalog_file)
    }
    switch(class(spec_data)[1], source_file = df_parse_sas_file(spec_data,
        spec_cat, encoding = encoding, catalog_encoding = catalog_encoding,
        cols_skip = cols_skip, n_max = n_max, rows_skip = skip,
        name_repair = .name_repair), source_raw = df_parse_sas_raw(spec_data,
        spec_cat, encoding = encoding, catalog_encoding = catalog_encoding,
        cols_skip = cols_skip, n_max = n_max, rows_skip = skip,
        name_repair = .name_repair), cli_abort("This kind of input is not
handled."))
}
<bytecode: 0x1189ed5f0>
<environment: namespace:haven>
```

👍 Export from SAS to csv

- I use SAS code to export only nececary data to csv
- I might use a prefiltering list exported from R to limit the data
- SAS Viya more flexible (I've heard) but not SAS 9.4
- Use `(set=1000)` etc for tests and then run over night
- csv-files exported to local SSD drive (and then synced back to storage folder)

ChatGPT

> Jag har exporterat en **csv-fil från R** som innehåller en lista med **löpnummer** på individer. Jag vill nu använda **SAS**. Först vill jag läsa in CSV-filen. Sedan vill jag **filtrera** ett dataset i SAS (vi kan kalla det pelle) som har en kolumn med samma löpnummer som i CSV-filen (jag vill bara ha med de fall där det finns en match). Därefter vill jag **exportera** resultatet till en **ny csv-fil**. Jag vill att detta ska ske på ett **effektivt och snabbt** sätt. Hjälp med med **SAS-syntax** för detta

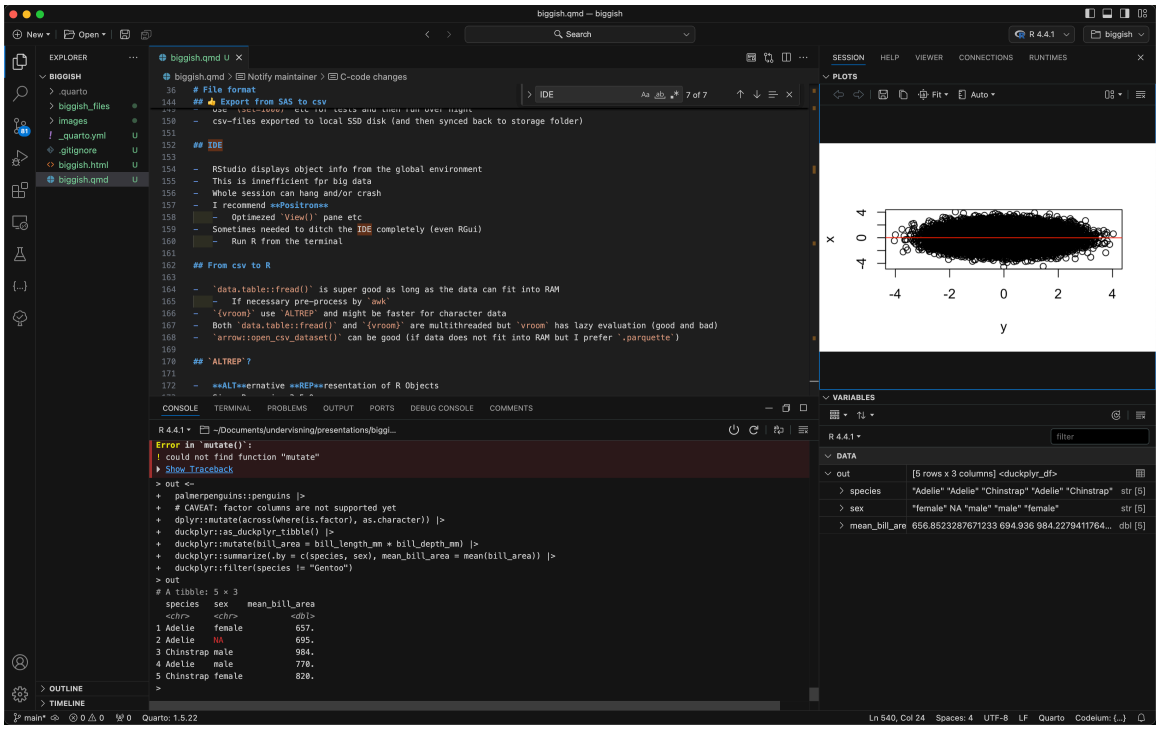
---

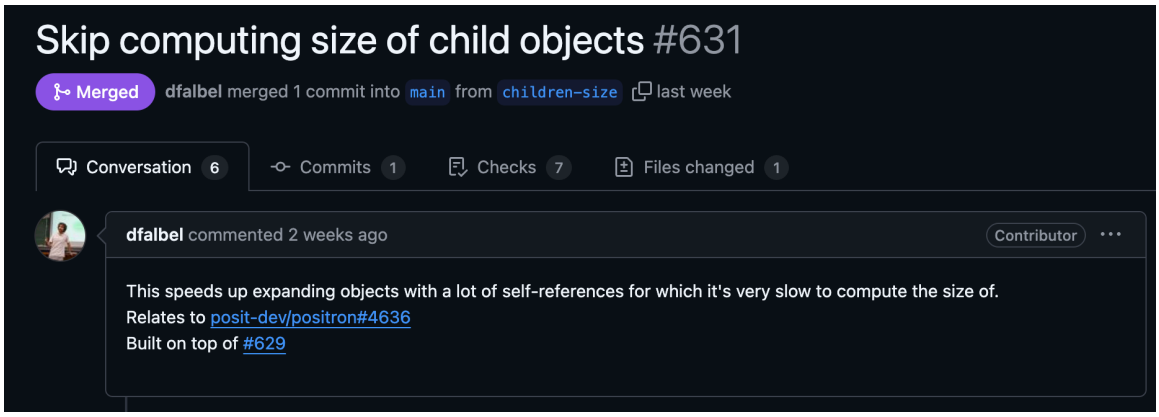
```
/* Läs in CSV-filen i SAS */
```

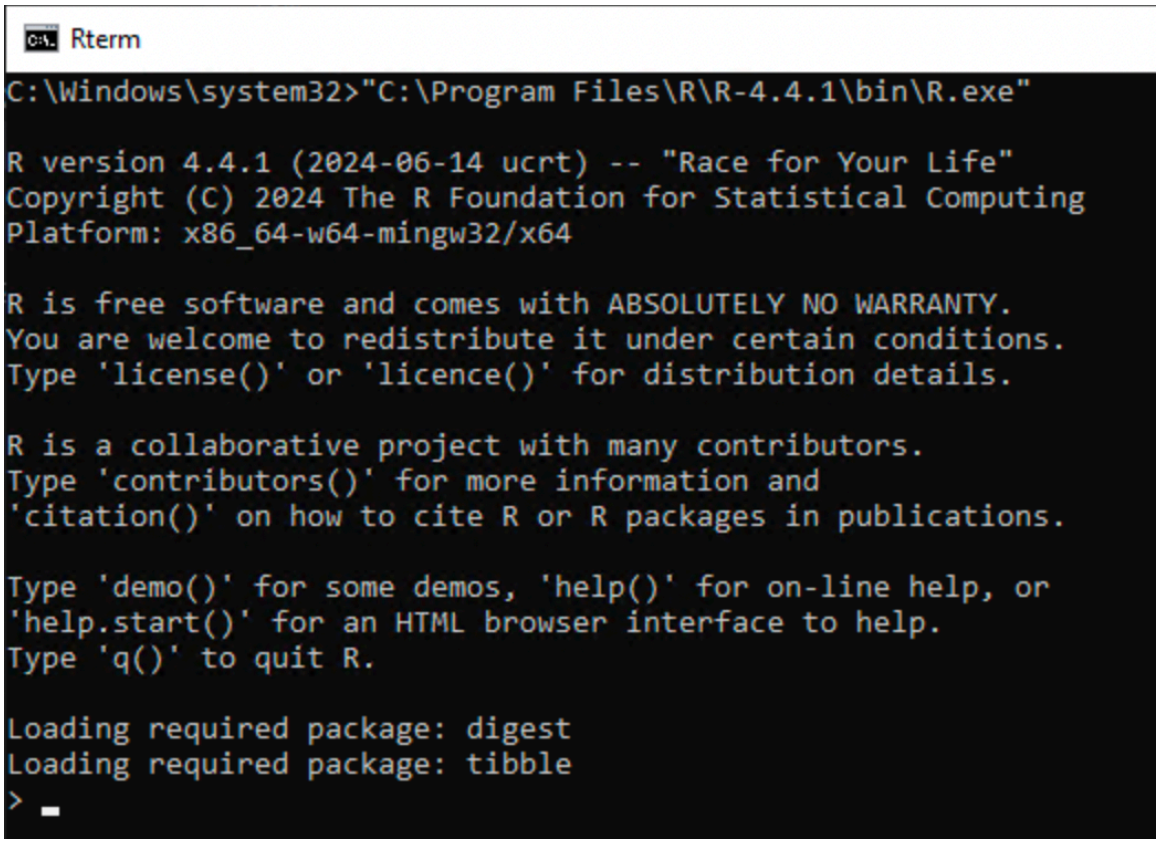
```
/* Läs in filen med löpnummer till ett SAS-dataset (vi kallar det `individer`)
*/
proc import datafile="din_sökväg_till_csv_filen.csv"
    out=individer>
    dbms=csv
    replace;
    getnames=yes; /* Sätt till no om csv-filen saknar rubriker */
run;

/* Om CSV-filen saknar rubrik kan du lägga till följande steg för att namnge
kolumnen manuellt */
data individer;
    set individer(rename=(var1=löpnummer)); /* Byt "var1" till korrekt
variabelnamn */
run;

/* Filtrera datasetet `pelle` baserat på löpnummer */

/* Vi använder en hash-lookup för att göra filtreringen effektiv. Hash-lookup
är snabbare än traditionella merge-metoder. */
data pelle_filtered;
    if _n_ = 1 then do;
        declare hash h(dataset: "individer");
        h.definekey("löpnummer");
        h.definedone();
    end;
    set pelle;
    if h.find() = 0; /* Endast matchande löpnummer */
run;

/* Exportera det filtrerade datasetet till en ny CSV-fil */

/* Använd proc export för att spara resultatet */
proc export data=pelle_filtered
    outfile="din_sökväg_till_ny_csv_fil.csv"
    dbms=csv
    replace;
run;

/* Effektivitetsoptimering:
Om både `individer` och `pelle` är mycket stora,
kan du sortera båda och använda data step merge
istället för hash-lookup. Hash-metoden är dock
generellt snabbare om det är många rader i båda
tabellerna. */
```

IDE

- RStudio displays object info from the global environment
- This is inefficient for big data
- Whole session can hang and/or crash
- I recommend Positron
  - Optimezed `View()` pane etc
  - Active and enthusiastic development
- Sometimes needed to ditch the IDE completely (even RGui)
  - Run R from the terminal

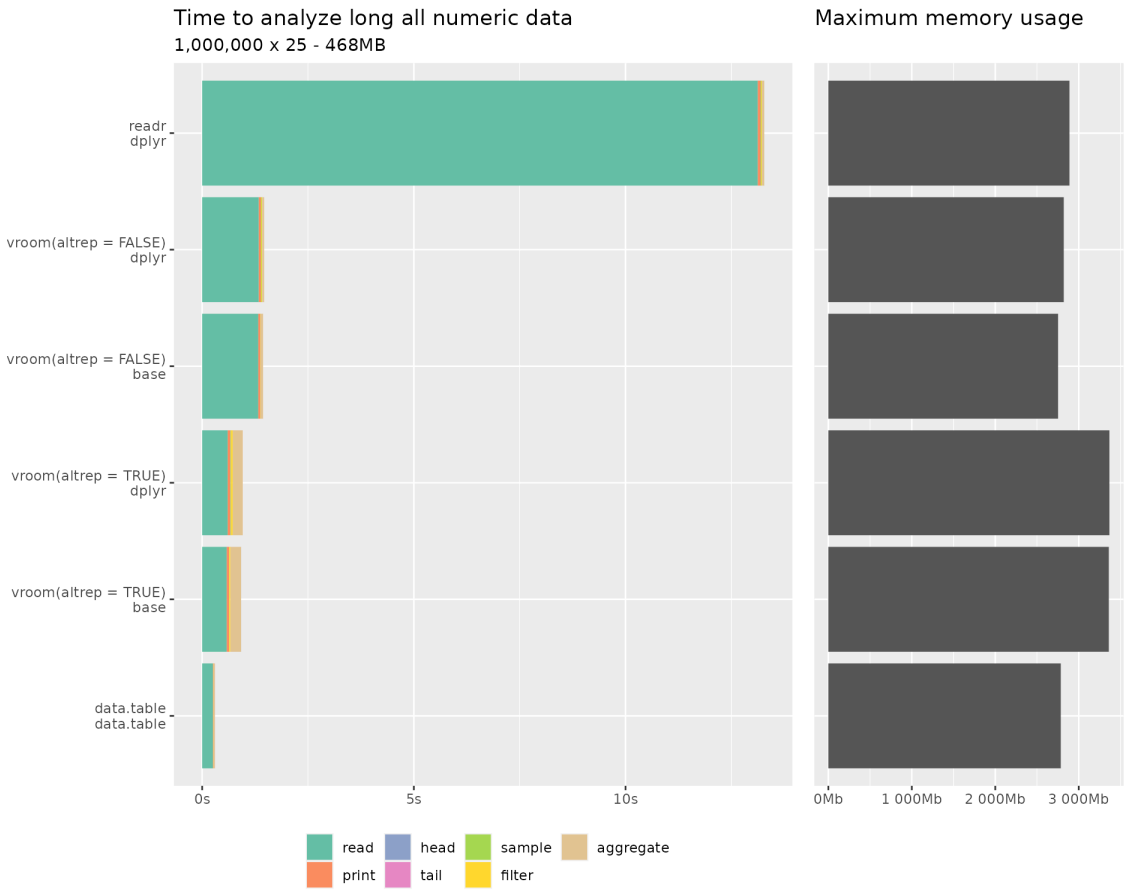






From csv to R

- `data.table::fread()` is super good as long as the data can fit into RAM
  - If necessary pre-process by `awk`
- `{vroom}` use `ALTREP` and might be faster for character data
- Both `data.table::fread()` and `{vroom}` are multithreaded but `vroom` has lazy evaluation (good and bad)
- `arrow::open_csv_dataset()` can be good (if data does not fit into RAM but I prefer `.parquette` if so)


Time to analyze long all numeric data
1,000,000 x 25 - 468MB
Maximum memory usage
readr
dplyr
vroom(altrep = FALSE)
dplyr
vroom(altrep = FALSE)
base
vroom(altrep = TRUE)
dplyr
vroom(altrep = TRUE)
base
data.table
data.table
0s
5s
10s
0Mb
1 000Mb
2 000Mb
3 000Mb
read
head
sample
aggregate
print
tail
filter

ALTREP?

- **ALT**ernative **REP**resentation of R Objects
- Since R version 3.5.0
- Applies to the C-level representation of R objects under the hood
- Used by some base R packages/functions + specialized packages

> allow vector data to be in a memory-mapped file; distributed, e.g. within Apache Spark or Hadoop; shared with other applications, e.g. with Apache Arrow; allow compact representation of arithmetic sequences; allow adding meta-data to objects; allow computations/ allocations to be deferred; support alternative representations of environments.

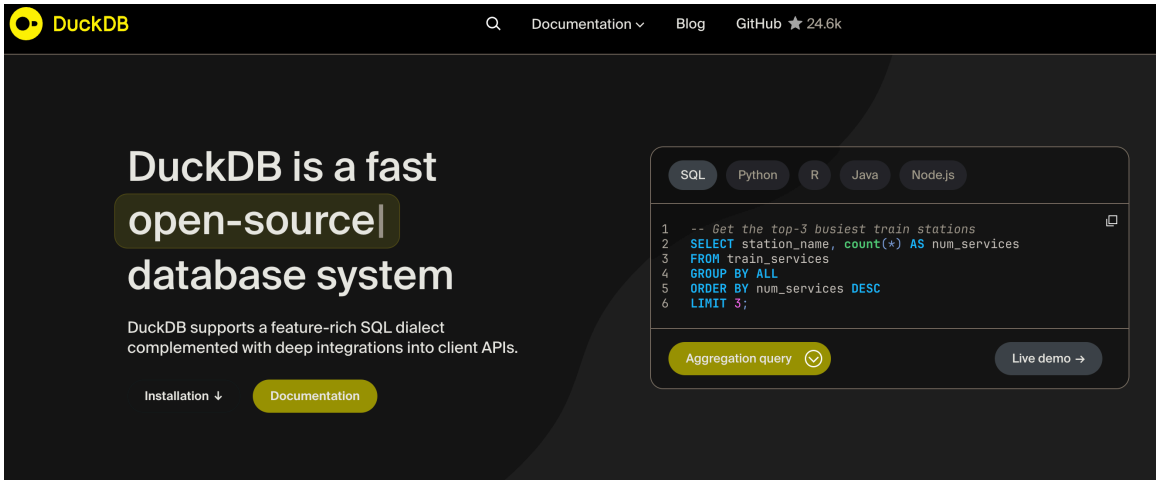
Database?

Database?

- I don't want to administer Microsoft SQL Server myself!
  - (They had R support for a couple of years but not any more)
- SQLight is easy to use …
- but (as SQL server) it is row based 😕
- Some complicated ones are column based 😀 (Snowflake, Google Big Query)
  - … but I don't want to administer those either …
- DuckDB comes for rescue!

! Indices

Never forget to set proper indices if needed!



DuckDB

- Free and open source (backed by foundation)
- Column based
- Very easy to set up
- Very efficient!
- Integration with tidyverse in R
- Can be used both with disk data and data in RAM
- … but I had some bad experience and now I am scared 😢

From CSV to Parquet

- CSV is not very nice 😕
  - No meta data
  - Sensitive to errors (missing values, decimal point, numbers with leading 0 etc, although `fread` can take care of this)
  - Encoding, BOM, whatever …
- Write to Parquet files
- I don't think it is possible to convert csv to parquett on disk (first read csv, then write parquette)
  - But it can be done in chunks (read portion of csv and write to individual parquet file which are later "combined")

Parquet

- From Apache Arrow
- Open format
- flat files with possible hierarchical structure by transparent structure of folders and filenames
- Integrates with DuckDB (which integrates with R/tidyverse)
- See example later!

R specific data files?

- I use `.qd`-files (previously `.qs`) to cache non-tabular or intermediate data objects

> {qs2}: a framework for efficient serialization … The qs2 format directly uses R serialization (via the R_Serialize/R_Unserialize C API) while improving underlying compression and disk IO patterns.

- `.RData` is terribly slow!
- `.rds` should now be much faster than before (serialized)
- `.fst` might be better in some cases but `.qd/.qs` is more generic (I think)

---

qs2

R-CMD-check passing | CRAN 0.1.2 | downloads 803/month | downloads 1274

qs2: a framework for efficient serialization

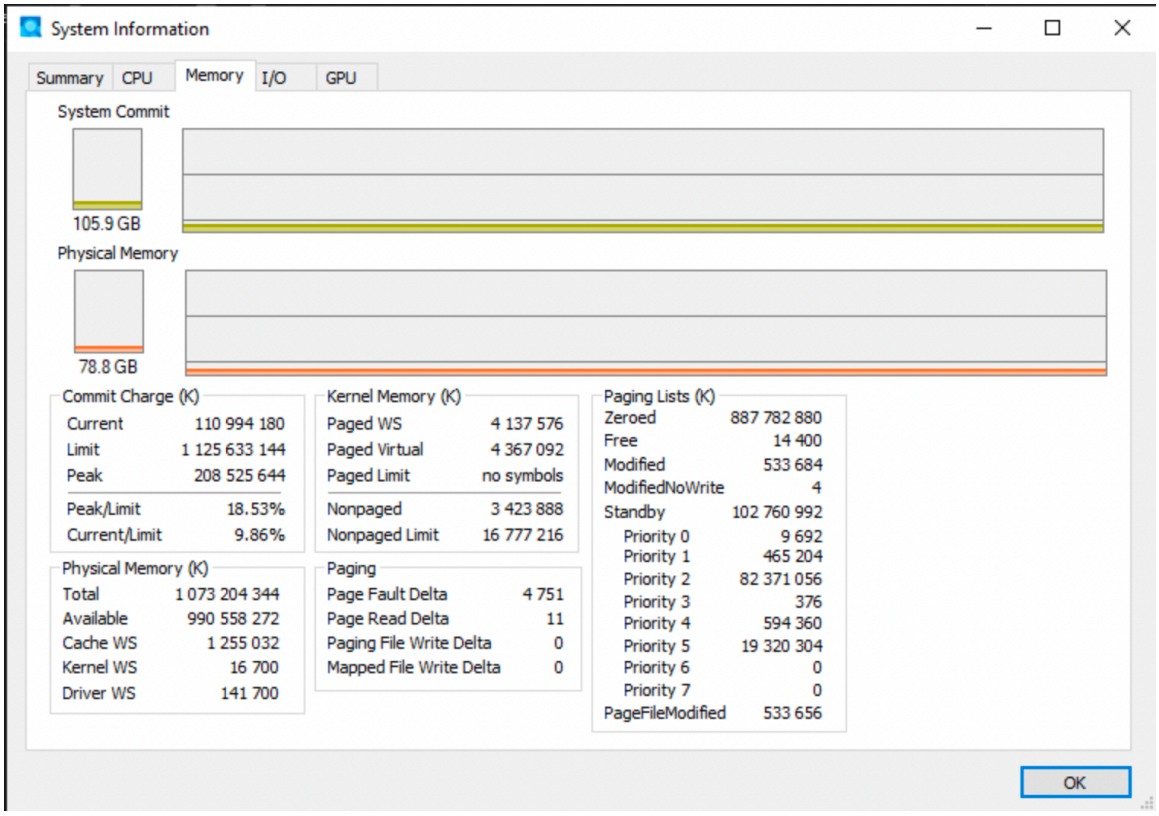
`qs2` is the successor to the `qs` package. The goal is to have reliable and fast performance for saving and loading objects in R.

The `qs2` format directly uses R serialization (via the `R_Serialize` / `R_Unserialize` C API) while improving underlying compression and disk IO patterns. If you are familiar with the `qs` package, the benefits and usage are the same.

```
qs_save(data, "myfile.qs2")
data <- qs_read("myfile.qs2")
```

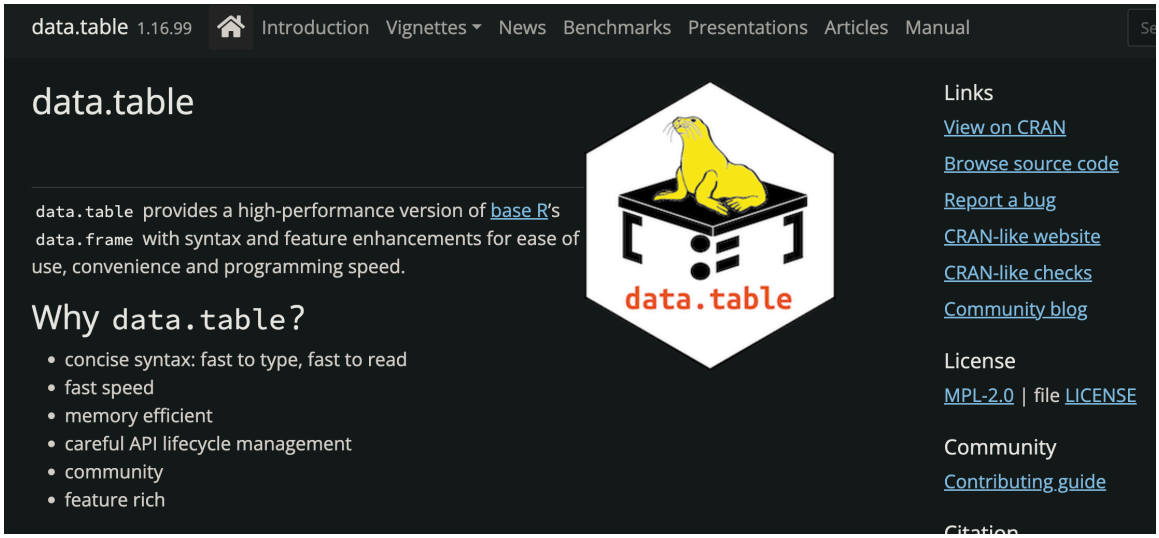
Use the file extension `qs2` to distinguish it from the original `qs` package. It is not compatible with the original `qs` format.

Data management



- Let's say data fits into RAM
- base R and tidyverse works ... but might be terribly slow ...

- {data.table}! 😀



{data.table}

- Share basic properties with data frames (`dt[i, j]; merge()`)
- Print method inspired by `{tibble}`
- Some old friends from `{reshape2}: melt(), dcast()`
- Has been around since 2006 (stable and well used, older than tidyverse)
- faster alternatives to base: `fifelse(), fcase(), forder(), fread(), fcoalesce(), as.IDate(), %chin%`
- C and GForce (some difficulties on Mac ...)
- integer based keys and indices (compare data bases)
- Reference semantics (no need to copy on modification)
- Very flexible and compact (but perhaps unusual) syntax

Example

```
library(data.table)
dt <- iris
setDT(dt)
setkey(iris)
iris
```

```
Key: <Sepal.Length, Sepal.Width, Petal.Length, Petal.Width, Species>
     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species
            <num>       <num>        <num>       <num>    <fctr>
  1:          4.3         3.0          1.1         0.1    setosa
  2:          4.4         2.9          1.4         0.2    setosa
  3:          4.4         3.0          1.3         0.2    setosa
```

```
  4:          4.4         3.2          1.3         0.2    setosa
  5:          4.5         2.3          1.3         0.3    setosa
 ---
146:          7.7         2.6          6.9         2.3 virginica
147:          7.7         2.8          6.7         2.0 virginica
148:          7.7         3.0          6.1         2.3 virginica
149:          7.7         3.8          6.7         2.2 virginica
150:          7.9         3.8          6.4         2.0 virginica
```

```
dt[, .N, Species]
```

```
      Species     N
       <fctr> <int>
1:     setosa    50
2: versicolor    50
3:  virginica    50
```

```
dt[, date := as.IDate(Sys.Date()) + .GRP, Species][]
```

```
     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species       date
            <num>       <num>        <num>       <num>    <fctr>     <IDat>
  1:          4.3         3.0          1.1         0.1    setosa 2026-02-26
  2:          4.4         2.9          1.4         0.2    setosa 2026-02-26
  3:          4.4         3.0          1.3         0.2    setosa 2026-02-26
  4:          4.4         3.2          1.3         0.2    setosa 2026-02-26
  5:          4.5         2.3          1.3         0.3    setosa 2026-02-26
 ---
146:          7.7         2.6          6.9         2.3 virginica 2026-02-28
147:          7.7         2.8          6.7         2.0 virginica 2026-02-28
148:          7.7         3.0          6.1         2.3 virginica 2026-02-28
149:          7.7         3.8          6.7         2.2 virginica 2026-02-28
150:          7.9         3.8          6.4         2.0 virginica 2026-02-28
```

```
dt[, names(.SD) := lapply(.SD, as.character), .SDcols = is.factor][]
```

```
     Sepal.Length Sepal.Width Petal.Length Petal.Width   Species       date
            <num>       <num>        <num>       <num>    <char>     <IDat>
  1:          4.3         3.0          1.1         0.1    setosa 2026-02-26
  2:          4.4         2.9          1.4         0.2    setosa 2026-02-26
  3:          4.4         3.0          1.3         0.2    setosa 2026-02-26
  4:          4.4         3.2          1.3         0.2    setosa 2026-02-26
  5:          4.5         2.3          1.3         0.3    setosa 2026-02-26
 ---
```

```
146:          7.7         2.6          6.9         2.3 virginica 2026-02-28
147:          7.7         2.8          6.7         2.0 virginica 2026-02-28
148:          7.7         3.0          6.1         2.3 virginica 2026-02-28
149:          7.7         3.8          6.7         2.2 virginica 2026-02-28
150:          7.9         3.8          6.4         2.0 virginica 2026-02-28
```

Tidyverse-like Data Manipulation built on *data.table*

- **tidytable**: A tidy interface to *data.table* that is *rlang* compatible. Quite comprehensive implementation of *dplyr*, *tidyr* and *purr* functions. Package uses a class *tidytable* that inherits from *data.table*. The `dt()` function makes *data.table* syntax pipeable (12 total dependencies).
- **dtplyr**: A tidy interface to *data.table* built around lazy evaluation i.e. users need to call `as.data.table()`, `as.data.frame()` or `as_tibble()` to access the results. Lazy evaluation holds the potential of generating more performant *data.table* code (20 dependencies).
- **tidyfst**: Tidy verbs for fast data manipulation. Covers *dplyr* and some *tidyr* functionality. Functions have `_dt` suffix and preserve *data.table* object. A cheatsheet is provided (7 dependencies).
- **tidyft**: Tidy verbs for fast data operations by reference. Best for big data manipulation on out of memory data using facilities provided by *fst* (7 dependencies).
- **tidyfast**: Fast tidying of data. Covers *tidyr* functionality, `dt_` prefix, preserves *data.table* object (2 dependencies).
- **maditr**: Fast data aggregation, modification, and filtering with pipes and *data.table*. Minimal implementation with functions `let()` and `take()` for most common data manipulation tasks. Also provides Excel-like lookup functions (2 dependencies).
- **table.express** also o builds *data.table* expressions from *dplyr* verbs, without executing them eagerly. Similar to *dtplyr* but less mature (17 dependencies).

  Notes: These packages are wrappers around *data.table* and do not introduce own compiled code.

If it doesn't fit?

- Connect to parquet file.
- Use DuckDB to process the data (while still on disk)
- Use `{dplyr}` to translate its syntax into SQL code
- Process data on disk
- Only read the results into memory

Example

```
library(tidyverse)
db1 <-
  arrow::open_dataset("file1.parquet") |>
  arrow::to_duckdb() |>
  select(col1:col3) |>
  filter(...) |>
  pivot_wider(..)

db2 <-
  arrow::open_dataset("file2.parquet") |>
  arrow::to_duckdb() |>
  count(x, y) |>
  filter(...)

df <-
  db1 |>
  left_join(db2) |>
  collect() # Only the resulting part should be collected!
```

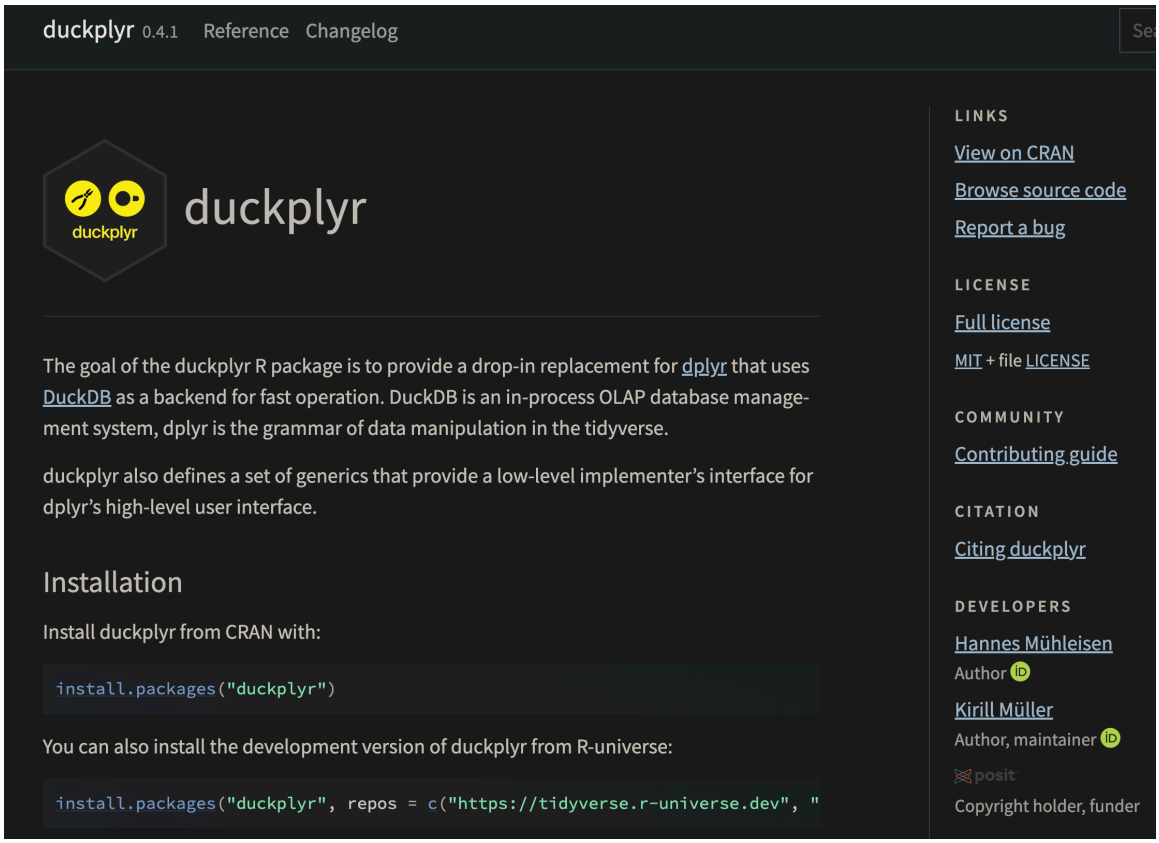
Pros 👍

- Lazy: nothing evaluated until it is actually needed (collected)
- Many basic functions are translated from R to SQL
- DuckDB is parallelized (`{data.table}` is mostly sequential)
- Familiar syntax from tidyverse

Cons 👎

- tidyverse -> SQL translation not perfect (non optimized)
- might be relatively slow if intermediate steps are not cached and needs to be performed repeatedly

---

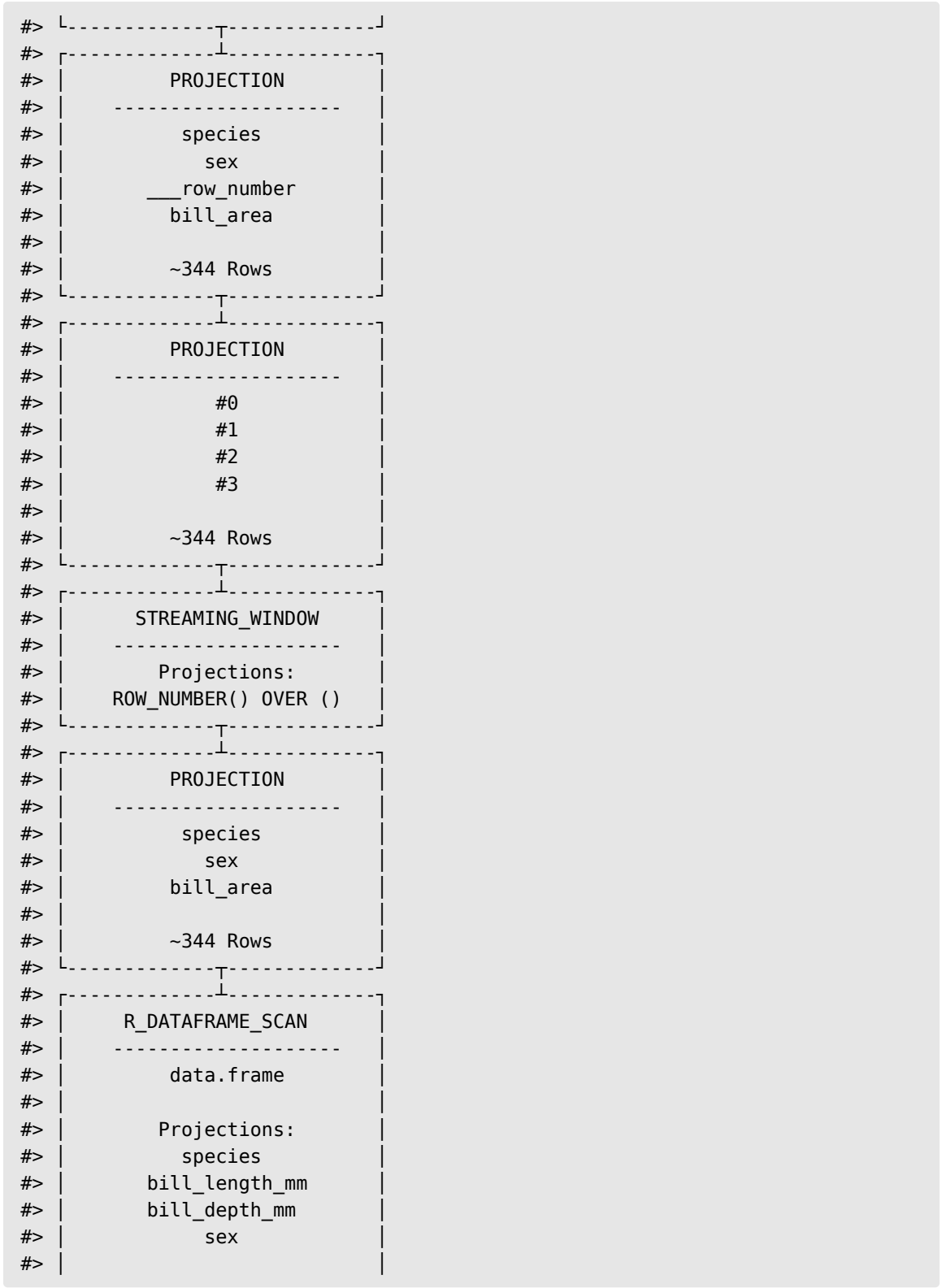


DuckPlyr

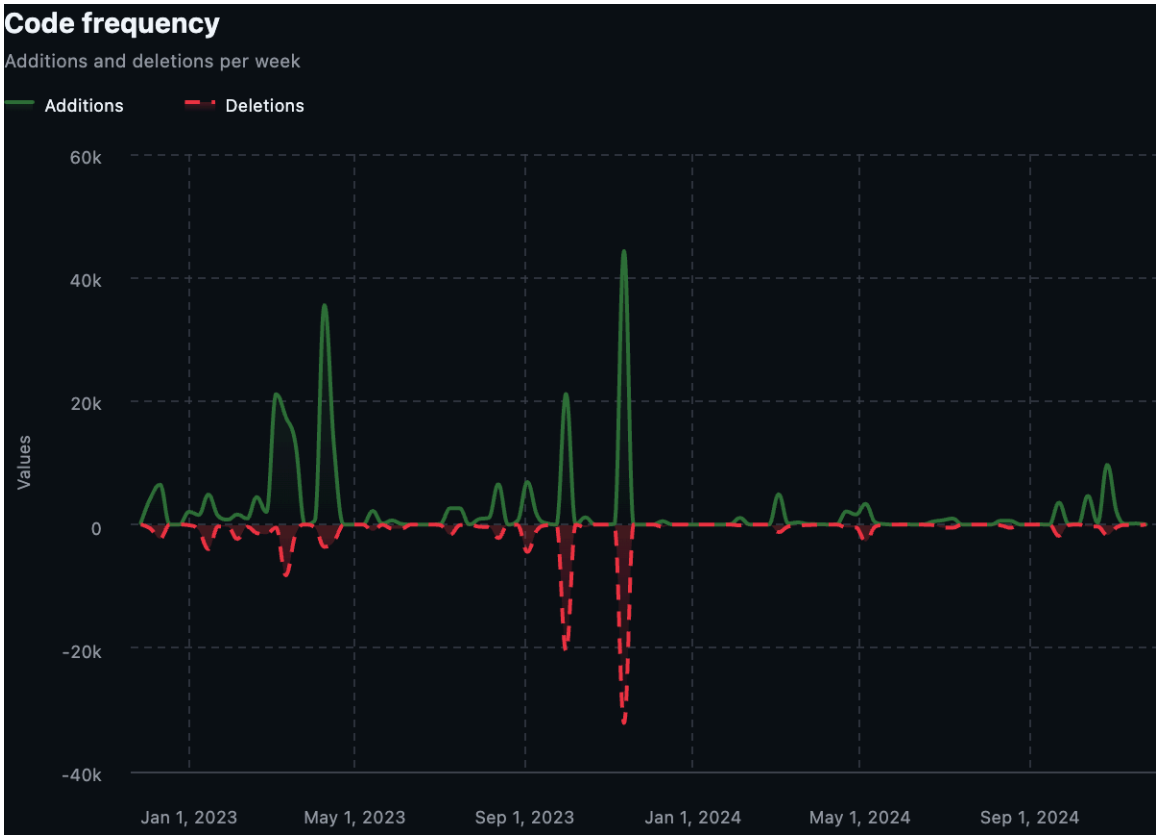
- New package with API-calls to DuckDB (no standard SQL code in between ... I think)
- Drop-in-replacement of {dplyr}
- Automatically fail-safe to collect and use standard {dplyr} if needed
- Uptimized engine to not touch more data than necessary!
- Although very limited functionality so far (but wait and see!)

```
out <-
  palmerpenguins::penguins %>%
  # CAVEAT: factor columns are not supported yet
  mutate(across(where(is.factor), as.character)) %>%
  duckplyr::as_duckplyr_tibble() %>%
  mutate(bill_area = bill_length_mm * bill_depth_mm) %>%
  summarize(.by = c(species, sex), mean_bill_area = mean(bill_area)) %>%
  filter(species != "Gentoo")

out %>%
  explain()
```

```
#> ┌---------------------------┐
#> │          ORDER_BY         │
#> │    --------------------   │
#> │      dataframe_42_42      │
#> │    42.___row_number ASC   │
#> └-------------┬-------------┘
#> ┌-------------┴-------------┐
#> │           FILTER          │
#> │    --------------------   │
#> │   "r_base::!="(species,   │
#> │         'Gentoo')         │
#> │                           │
#> │          ~34 Rows         │
#> └-------------┬-------------┘
#> ┌-------------┴-------------┐
#> │         PROJECTION        │
#> │    --------------------   │
#> │             #0            │
#> │             #1            │
#> │             #2            │
#> │             #3            │
#> │                           │
#> │         ~172 Rows         │
#> └-------------┬-------------┘
#> ┌-------------┴-------------┐
#> │      STREAMING_WINDOW     │
#> │    --------------------   │
#> │        Projections:       │
#> │    ROW_NUMBER() OVER ()   │
#> └-------------┬-------------┘
#> ┌-------------┴-------------┐
#> │          ORDER_BY         │
#> │    --------------------   │
#> │      dataframe_42_42      │
#> │    42.___row_number ASC   │
#> └-------------┬-------------┘
#> ┌-------------┴-------------┐
#> │       HASH_GROUP_BY       │
#> │    --------------------   │
#> │          Groups:          │
#> │             #0            │
#> │             #1            │
#> │                           │
#> │        Aggregates:        │
#> │          min(#2)          │
#> │          mean(#3)         │
#> │                           │
#> │         ~172 Rows         │
```


PROJECTION
species
sex
___row_number
bill_area
~344 Rows
PROJECTION
#0
#1
#2
#3
~344 Rows
STREAMING_WINDOW
Projections:
ROW_NUMBER() OVER ()
PROJECTION
species
sex
bill_area
~344 Rows
R_DATAFRAME_SCAN
data.frame
Projections:
species
bill_length_mm
bill_depth_mm
sex

```
#> │          ~344 Rows          │
#> └─────────────────────────────┘
```

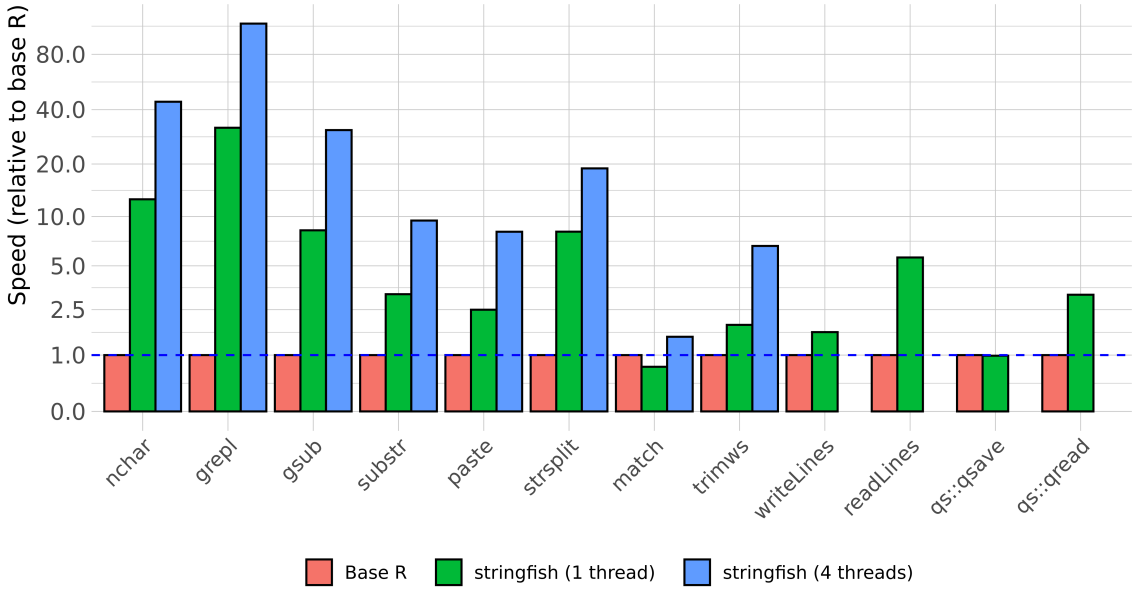


Dates

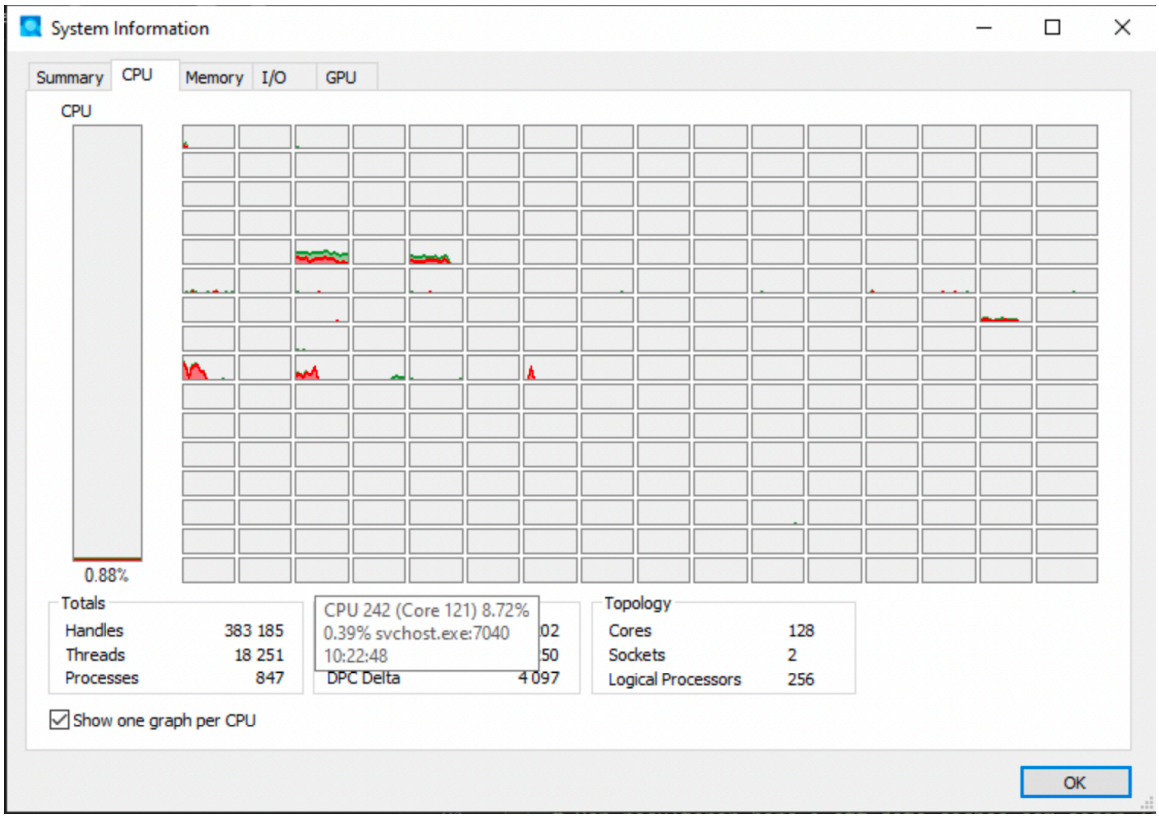
- Use `as.IDate()` instead of `as.Date()` if using `{data.table}`. Integer based.
- Use `{anytime}` (or maybe `{fasttime}`) if conversion is needed (`ALTREP`)

String manipulation

- Character manipulation in base R can be quite slow (does not use `ALTREP` and only single core by default)
  - For example regular expressions (`grep()`, `gsub()` etc)
- `{stringfish}` use multithreads and ALTREP - much faster!



Paralellization



Paralellization

- R is single threaded by default
- Some packages and functions may use multithreading by default
- Some functions might have arguments such as `nthreads`, `ncores` etc
- To handle multicore/thread computations can be (very) difficult (I've heard)
- Easy if data can be divided and conquered (split-apply-combine)
- `{futureverse}` is my favorite (Henrik Bengtsson)
  - `{furrr}` is a a drop-in replacement for `{purrr}`

{furrr}

```
# How many cores do we have?
parallel::detectCores()
```

```
[1] 12
```

```
library(furrr)
```

```
Loading required package: future
```

```
# Save at least one for OS etc
plan(multisession, workers = parallel::detectCores() - 1)

1:10 |>
  future_map(rnorm, n = 10, .options = furrr_options(seed = 123)) |>
  future_map_dbl(mean)
```

```
 [1] 1.180279 2.140442 2.909823 3.692207 5.058100 6.653926 7.065630 7.960713
 [9] 9.105674 9.766827
```

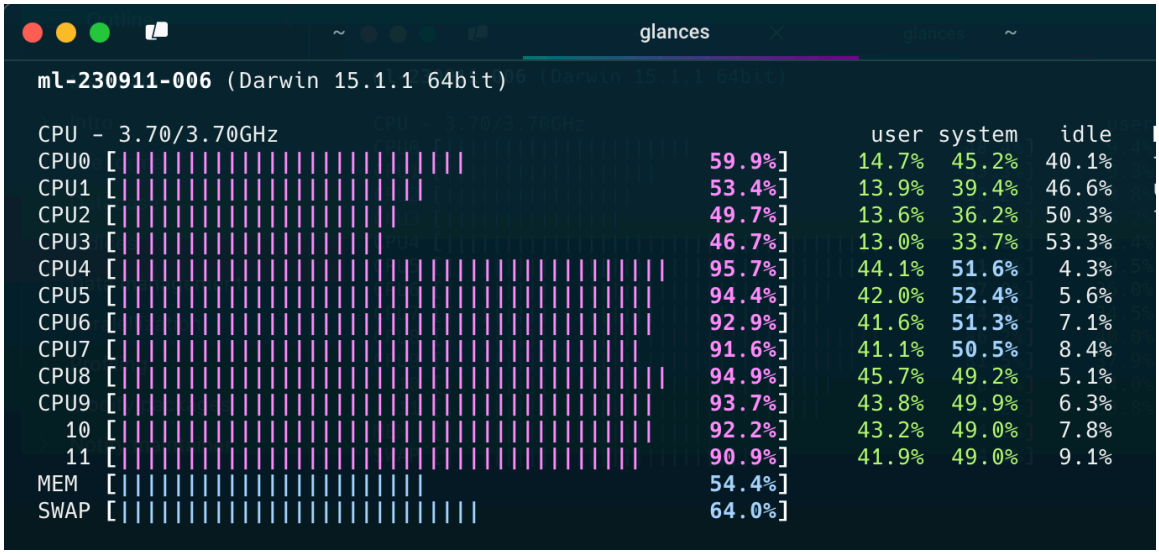
{mirai}

- Suposed to be up to 1,000 times "faster" compared to `{furrr}`(according to some metric ...)
- should not block the main process while executing (haven't tried)

> Designed for simplicity, a 'mirai' evaluates an R expression asynchronously in a parallel process, locally or distributed over the network, with the result automatically available upon completion.

```
library(mirai)
daemons(11, dispatcher = "thread") # "thread" still experimental
```

```r
# data.table with 1 billion rows:
dt <- data.table(x = seq_len(1e9), y = rnorm(1e9))
# row sums (row wise )
mirai_map(dt, sum)[.flat]
```



Note

- Data must be copied for each paralell worker
- If all data is needed for each worker, it will be multiplied in memory
  - This might not be possible for big data
  - It also takes time
- Hence, there is no guarantee that the over all time will decrease
- Often a tradeof between no of workers/threads/deamons/cores and efficiency etc
- "Standard" computations in `{data.table}` is optimized for single core
- Message handling and progress bar etc is complicated

Caching

- I use `{ProjectTemplate}` to organize R projects into different steps
- It has an inbuilt cache mechanism through `ProjectTemplate::cache()`
- I have my own cache-functions in my `.Rprofile`

```r
# OBS kollar inte att det verkligen finns cache-folder etc
cache_dt <- function(x, name = deparse(substitute(x)), nthreads = 5) {
  qs2::qd_save(x, glue::glue("cache/{name}.qd"), nthreads = nthreads)
  invisible(x)
}
```

```r
decache_dt <- function(name, nthreads = 5) {
  x <- qs2::qd_read(glue::glue("cache/{name}.qd"), TRUE, nthreads = nthreads)
  if (data.table::is.data.table(x)) {
    data.table::setDT(x)
  }
  assign(name, x, envir = .GlobalEnv)
  invisible(x)
}
```

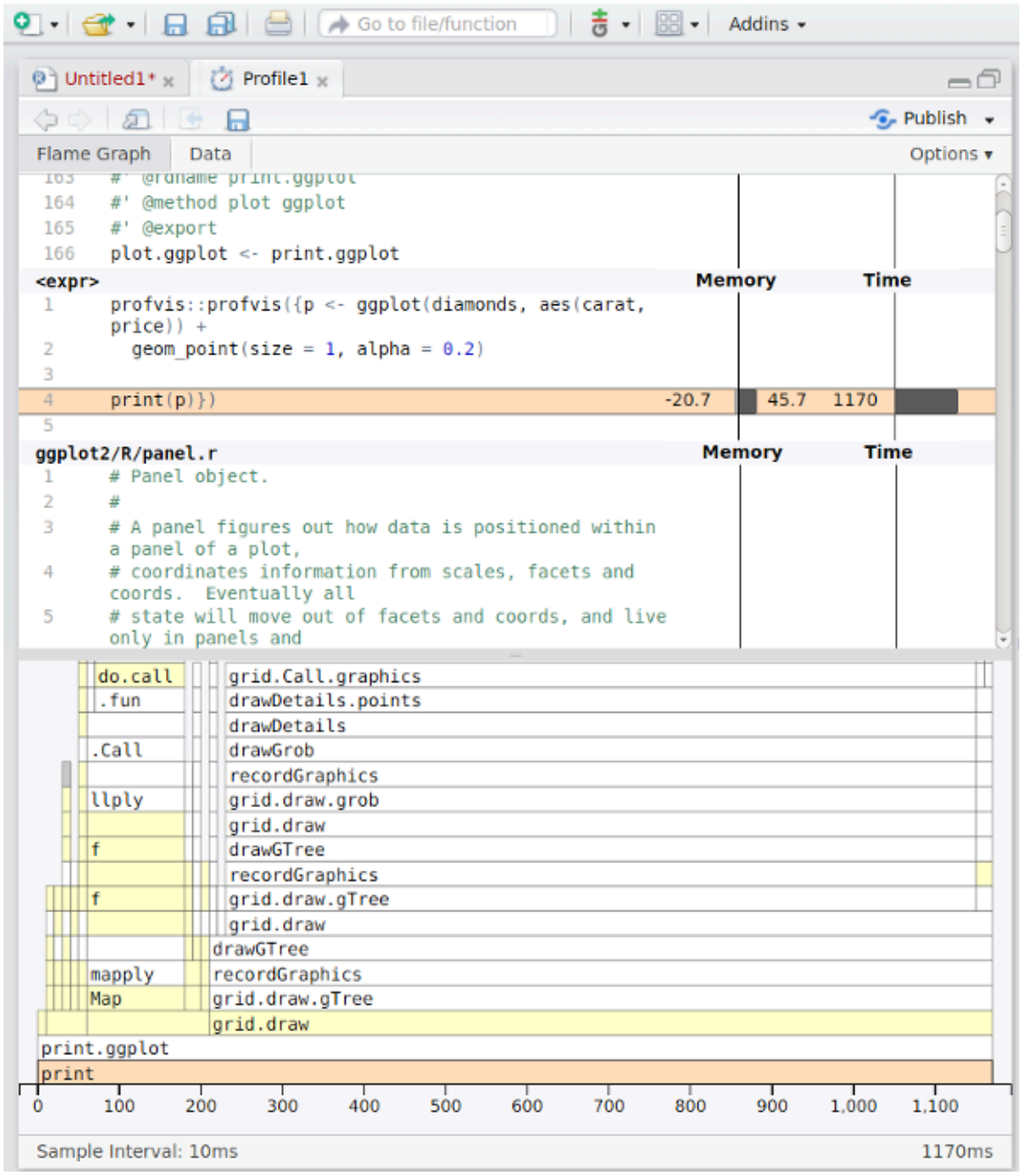
Intermediate caching

- For long running function calls (hours, days)
- Divide task into smaller pieces and execute each part in paralell if feasable
- Use wrapper function with some index/grouping variable to indicate which part is currently processed
- Cache intermediate object as soon as possible (with name based on the grouping variable)
- If process is disrupted and must be restarted, parts which are already executed are just retrieved from from cache

```r
cache_wrapper <- function(dt_part, name, ...) {
  if (fs::file_exists(glue::glue("cache/{name}.qd"))) {
    decache_dt(name)
  } else {
    x <- do_what_should_be_done()
    cache_dt(x, name, ...)
    x
  }
}
```

Profiling

- First version of R script might be inneficient
- Optimazation is difficult
- Might be suboptimal if made ad hoc
- `{profvis}` visualise time and memory usage for each step
- improve the most important step first
  - For example change `{base}` and `{tidyverse}` code to `{data.table}`
  - set keys
  - Efficient handling of dates and strings
- itterate
- Integrated in RStudio (but not yet Positron)



Modify packages

Modify packages

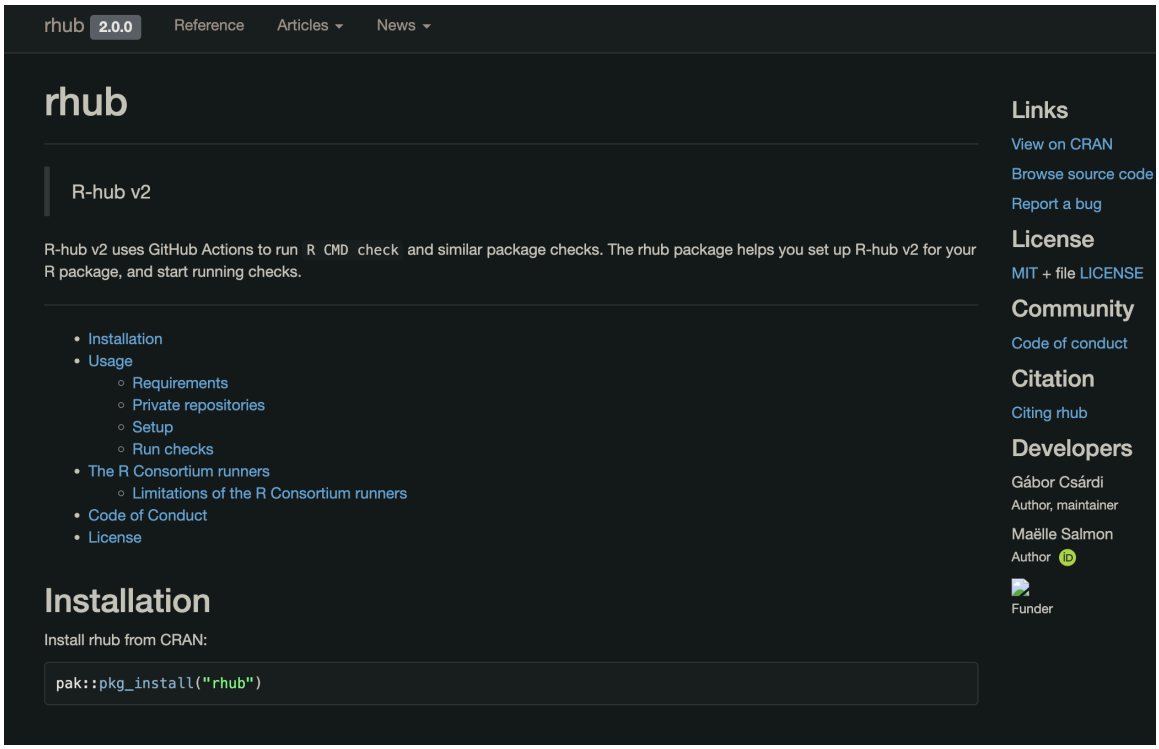
- Functions from packages might be inefficient
- Profile those as well
- Clone package from GitHub if available or download source code from CRAN
- Improve

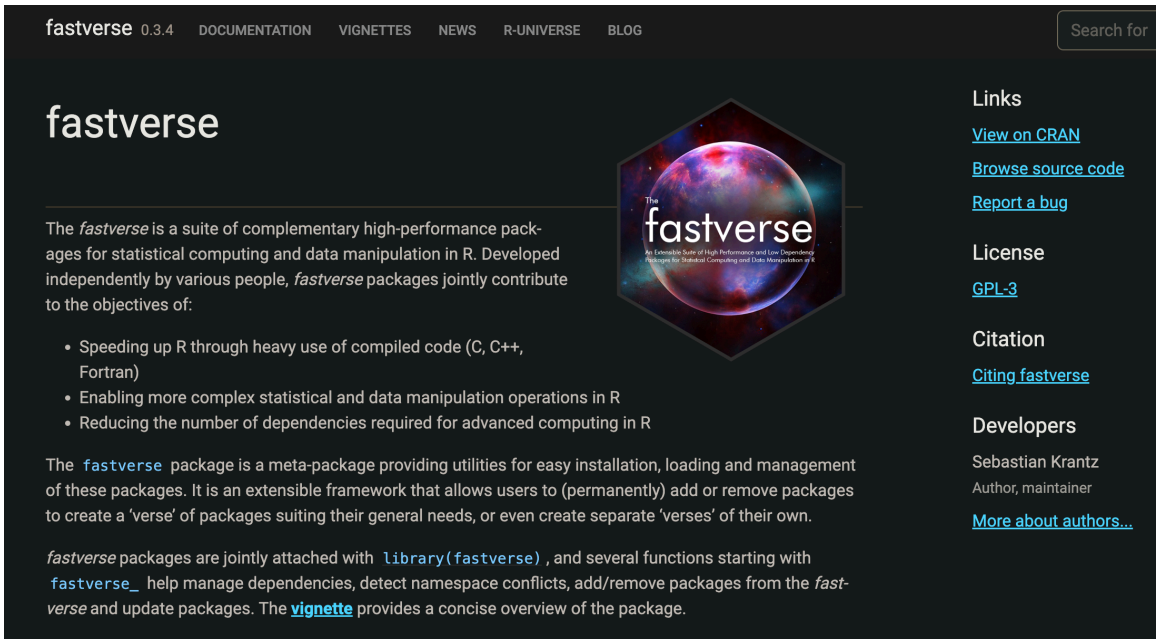
- Just load the modified function in global environment
  - might need to modify internal calls to `:::`-accesed functions
- Or rebuild/install the package if more convenient

> ! Notify maintainer
>
> If you find ways to improve a package, the maintainer might be eager to hear! Suggest pull request or open GitHub issue!

C-code changes

- If the package use C-code it is probably already very fast
- If not, and you can fix it, the package must be re-compiled
- OS dependent
- Might be tricky if using restricted environments (TRE?)
- Can still be done by rhub (might need to change maintainer-e-mail temporarily)





Bibliography